Gas Discharge Arc Lamp Manufacturers — 528nm SPR / PGM-REE Processing
Specification: Dielectric breakdown gas discharge apparatus — gaseous medium in dielectric housing sustaining electrical arc discharge — primary emission 528 nm (green) for Au/Ag SPR excitation — continuous or pulsed UV-Vis-NIR across PGM and REE absorption bands.

Technology Overview

Technology Gas Medium Housing 528nm? UV Coverage NIR Coverage
Xe Short Arc Xenon Quartz (SiO■) Continuum ✓ < 200 nm to 2500 nm
Hg-Xe Arc Hg + Xe Quartz + 546nm line 185 nm+ to 2500 nm
Kr Arc Krypton Quartz Continuum ✓ < 200 nm to 1000 nm
LDLS (laser Xe) Xenon Quartz dielectric Continuum ✓ 170 nm to 2100 nm
CERMAX Xenon Al■O■ ceramic Continuum ✓ < 250 nm to 1000 nm

DBD Xe/Hg/Kr Dielectric glass Narrow band VUV only limited
Microwave plasma Sulfur/Xe Quartz ~520 nm pk ✓ visible limited
Sulfur plasma S vapor Quartz 520 nm peak ✓ 430–750 nm limited

★★★★★ BEST MATCH — Energetiq Technology (Hamamatsu group)
Product: LDLS EQ-99XFC / EQ-77
Laser-Driven Light Source: a CW laser sustains a Xe plasma inside a sealed quartz dielectric bulb — electrodeless, no electrode erosion. Delivers 170–2100 nm continuous output, covering every PGM and REE absorption band. ~10× higher radiance than conventional Xe arcs. Fiber-coupled output directly to reaction vessel.
Website: https://www.energetiq.com/ldls-laser-driven-light-source
Parameter EQ-99XFC EQ-77
Spectral range 170–2100 nm 200–1100 nm
Output power > 1 mW/nm @ 400 nm > 0.5 mW/nm
Stability < 0.2% RMS < 0.5% RMS
Dielectric housing Fused silica (SiO■) Fused silica
Discharge type Laser-sustained Xe plasma Laser-sustained
Arc size < 100 µm < 100 µm
Fiber output SMA-905 or custom SMA-905

★★★★★ LITERAL SPEC MATCH — Excelitas Technologies — CERMAX

Product: PE300BF / PE75BF / PE150BF
CERMAX = Ceramic Maximum — Al■O■ ceramic reflector IS the dielectric housing. Xe arc burns inside a parabolic ceramic
cavity, producing collimated UV-Vis-NIR output. Long life (1000+ hrs), cold-restrike capable. Widely used in photodynamic
therapy, spectroscopy, and photocatalytic research. Available 75W–1600W.

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Website: https://www.excelitas.com/product-category/cermax-xenon-arc-lamps
Model Power Life Spectral range Color temp
PE75BF 75 W 500 hr 240–2500 nm 5500 K
PE150BF 150 W 1000 hr 240–2500 nm 5800 K
PE300BF 300 W 1000 hr 240–2500 nm 6000 K
PE1000 1000 W 500 hr 230–2500 nm 6200 K

★★★★ TOP PICK FOR 528nm — Sulfur Plasma / Topanga Technologies (Luxim) Product: LiFi Sulfur Plasma Lamp
Sulfur vapor microwave plasma in quartz dielectric bulb. Natural emission peaks at ~520 nm — closest of any discharge lamp to your 528 nm SPR target without bandpass filtering. 430–750 nm visible band. 1000W microwave drive, electrodeless, 50,000+ hr life. Heraeus and LumaCept are alternative suppliers of sulfur plasma systems.
Website: https://www.topangatech.com
Parameter Value
Peak emission ~520 nm (natural)
Spectral range 430–750 nm
Drive frequency 2.45 GHz microwave
Housing Fused quartz dielectric bulb
Lifetime 50,000+ hours
Input power 1000–1400 W

★★★★ — Newport / MKS Instruments — Oriel Series

Product: Model 66902 / 69911 / 67005 Xe Arc
Industry-standard Xe short arc systems. Quartz envelope xenon arc, 150W–1000W, UV-Vis-NIR continuum 200–2500 nm. Paired with Oriel monochromators or bandpass filters for wavelength selection. The 66902 (150W) is commonly used in photocatalysis research exactly matching this PGM/REE processing application.
Website: https://www.newport.com/c/arc-lamp-sources
Model Power Range Use case
66902 150 W 200–2500 nm Photocatalysis, SPR
69911 300 W 200–2500 nm High irradiance
67005 500 W 200–2500 nm Large beam area
Hg-Xe 66142 200 W 185–2500 nm UV-enhanced + 546nm line

★★★★ — Hamamatsu Photonics
Product: L10671 / L11035 / LC8 Series
Hamamatsu offers Xe short arc lamps (continuous and flash) plus their Lightningcure LC8 UV spot light source. The L10671 is
a compact fiber-coupled Xe lamp widely used in optical spectroscopy. Flash Xe lamps (L9455 series) provide pulsed µs–ms
pulses for time-resolved PGM photoluminescence measurements.

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Website: https://www.hamamatsu.com/us/en/product/light-sources/xenon-flash-lamps.html
Model Type Range Output

L10671 CW Xe 200–2000 nm 150 W

L11035 CW Xe 200–2000 nm 75 W compact
L9455 Flash Xe 200–1000 nm 0.1–10 ms pulse
LC8 UV spot 240–400 nm 200 mW/cm²

★★★ — ams OSRAM — XBO Series
Product: XBO 75W/2 / XBO 150W/4 / XBO 300W
XBO short arc xenon lamps — ultra-high pressure, small arc gap (< 1mm), extremely high luminance. Used in cinema projectors and scientific instruments requiring small source size (high etendue). Available with UV-transmitting quartz for deep UV output down to 230 nm — covers Pt (265 nm), Os, and Ir primary lines.
Website: https://www.ams-osram.com/products/lamps/xenon-short-arc-lamps
Model Power Arc gap Color temp
XBO 75W/2 75 W 0.8 mm 6200 K
XBO 150W/4 150 W 1.2 mm 6200 K
XBO 300W/4 300 W 2.0 mm 6000 K
XBO 1000W/HS 1000 W 5.5 mm 6100 K

★★★ — Ushio America
Product: UXL / SX Series Xenon Arc
Ushio produces high-quality Xe short arc and Hg-Xe lamps. The UXL-75XE is a workhorse for spectroscopy. Ushio also manufactures specialty Xe lamps with ceramic holders for harsh environments. Their SX series includes lamps optimized for UV curing and photocatalytic reactor applications.
Website: https://www.ushio.com/products/arc-lamps/
Model Power Type Range

UXL-75XE 75 W Xe arc 230–2500 nm
UXL-300D 300 W Xe arc 230–2500 nm
UXL-500D 500 W Xe arc 230–2500 nm

SX-UID75XAM 75 W Xe spot 240–700 nm

★★★ — Resonance Ltd
Product: Microwave-Excited Electrodeless Lamps
Canadian manufacturer specializing in electrodeless microwave-excited discharge lamps for analytical spectroscopy. Gas-fill selectable: Xe, Kr, Ar, or Hg — each giving different spectral line patterns. Quartz dielectric bulb, no electrodes (electrodeless discharge), 2.45 GHz microwave driver. Pulsed or CW modes.
Website: https://www.resonance.on.ca
Fill gas Key emission lines

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Xenon 823, 828, 881, 895, 979 nm + continuum
Krypton 758, 760, 769, 811 nm sharp lines
Mercury 185, 254, 365, 405, 436, 546, 578 nm
Argon 695, 763, 801, 811, 842, 912 nm

★★★ — Edinburgh Instruments
Product: Xe900 / µF920H Xe Flashlamp
Edinburgh Instruments produces pulsed Xe flashlamps and steady-state Xe sources for fluorescence spectroscopy. The µF920H offers sub-microsecond pulses — ideal for time-resolved luminescence of Eu³■, Tb³■, Nd³■ REE ions during separation monitoring. 200–900 nm range, quartz housing.
Website: https://www.edinst.com/products/light-sources/
Model Type Pulse width Range
Xe900 CW Xe arc continuous 200–900 nm

µF920H Flash Xe < 1 µs 200–900 nm
EPLED Pulsed LED < 100 ps 265–980 nm

★★ — Heraeus Noblelight
Product: Xe / Kr / DBD Specialty Lamps
Heraeus manufactures specialty discharge lamps including dielectric barrier discharge (DBD) excimer lamps emitting at narrow UV bands (172 nm Xe■*, 222 nm KrCl*, 308 nm XeCl*). These are true DBD lamps — dielectric on both electrodes, no arc, silent discharge — useful for Pt/Os/Ir primary UV lines. Also produces broad-spectrum Xe flash lamps for photocatalytic reactor applications.
Website: https://www.heraeus.com/en/hng/products_and_solutions/uv_curing/xenon_flash_lamps.html
Type Wavelength Application
Xe■* excimer 172 nm VUV photolysis, Os/Ir
KrCl* excimer 222 nm Ir line region
XeCl* excimer 308 nm Rh photocatalysis
Xe flash 200–2500 nm Broad photocatalysis

Spectral Coverage Matrix vs PGM/REE Targets
Target λ (nm) Energetiq LDLS CERMAX Xe Sulfur Plasma Hg-Xe Arc DBD Excimer

Au SPR 520–528 ✓ ✓ ★ Peak here ✓ —

Ag SPR 400–420 ✓ ✓ — ✓ —
Pd 340 ✓ ✓ — ✓ XeCl 308nm~
Rh 343 ✓ ✓ — ✓ XeCl 308nm~
Pt 265 ✓ ✓ — ✓ KrCl 222~
Os 226 ✓ ✓ — ✓ Xe■* 172~

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Ir 224 ✓ ✓ — ✓ Xe■* 172~

Eu³■ 394 ✓ ✓ — ✓ —

Tb³■ 365 ✓ ✓ — ✓ —

Er³■ 660 ✓ ✓ ✓ ✓ —

Nd³■ 808 ✓ ✓ — ✓ —

Yb³■ 980 ✓ ✓ — ✓ —

Priority Purchase Path

Goal Product Manufacturer URL Est. Cost

Best overall UV-Vis-NIR EQ-99XFC LDLS Energetiq <link href="https://www.energetiq.com" color="#0D47A1"><u>energetiq. $15k–25k

528nm natural peak Sulfur plasma lamp Topanga / Luxim <link href="https://www.topangatech.com" color="#0D47A1"><u>topang $3k–8k

Budget Xe arc + filter PE150BF CERMAX Excelitas <link href="https://www.excelitas.com/product-category/cermax-xenon-a $500–2k

Pulsed REE time-resolve µF920H Flashlamp Edinburgh <link href="https://www.edinst.com/products/light-sources/" color="#0D4 $2k–5k

Deep UV (Pt/Os/Ir) Xe■* 172nm DBD Heraeus <link href="https://www.heraeus.com" color="#0D47A1"><u>heraeus.co $1k–3k

Cross-referenced against STARE Research Division PDF "High-Capacity LED Systems for Metallurgical Photonic Processing" June 25, 2026.
All hyperlinks are active — tap/click to open manufacturer product pages.

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